PRODUCT USER MANUAL

For OMI_CIRCULATION_MOC_BLKSEA_area_averaged_mean

Issue: 1.0

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Approval Date: NOVEMBER 2024

MERCATOR OCEAN
INTERNATIONAL

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2024/11/26	All	Initial version	L. Lima, E. Jansen, M. Ilicak, R. Lecci	

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) OMI_CIRCULATION_MOC_BLKSEA_area_averaged_mean. It contains 1 OMI:

- omi_circulation_moc_blksea_area_averaged_mean

The OMI provides time series of Meridional Overturning Circulation Strength (MOC) area averaged time series calculated in the period 1993 to -1Y in the Black Sea.
It contains 1D (time dimension) MOC information computed from the Black Sea Reanalysis BLKSEA_MULTIYEAR_PHY_007_004 (Ilicak et al., 2022). The MOC is calculated by summing the meridional transport in bins of density. The Black Sea MOC indicator considers the maximum value of the MOC over the basin for a density between 22.45 and 23.85 kg/m^3. This corresponds approximately to a depth interval between 25 and 80m. To understand the overturning circulation of the Black Sea, we compute the residual meridional overturning circulation in density space. Residual overturning as a function of latitude (y) and density ($\bar{\sigma}$) bins can be computed as follows:

$$\psi^*(y,\bar{\sigma}) = -\frac{1}{T}\int_{t_0}^{t_1}\int_{x_{B1}}^{x_{B2}}\int_{-H}^{0}\mathcal{H}[\bar{\sigma}-\sigma(x,y,z,t)]\times v(x,y,z,t)dzdxdt,$$

where $\mathcal{H}$ is the Heaviside function and v is the meridional velocity. We used 100 σ_2 (potential density anomaly with reference pressure of 2000 dbar) density bins to remap the mass flux fields.

OMI_CIRCULATION_MOC_BLKSEA_area_averaged_mean contains annual mean time series of MOC. Units are in m^3 s^{-1}.

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is described in the Quality Information Document (QUID) CMEMS-OMI-QUID-BLK-CIRCULATION-MOC-BLKSEA.pdf.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CIRCULATION_MOC_BLKSEA_area_averaged_mean
Geographical coverage	Black Sea
Variables	ocean_meridional_overturning_streamfunction
Available time series	From 1993 to -1Y
Temporal resolution	Yearly means
Horizontal resolution	1/40
Integration depth (for example)	N/A
Format	NetCDF4

b) Production System Description

Data extracted from the Copernicus Marine Black Sea Reanalysis at 1/40 degree resolution provide the velocity fields which are used to compute the series of the Black Sea MOC. The annual mean time series starts on 1993 and ends -1Y, each yearly mean is centred on the 30th of June. This product includes annual mean files of MOC temporal series for the Black Sea.

c) Dataset details

OMI_CIRCULATION_MOC_BLKSEA_area_averaged_mean contains the following variables:
MOC [m3 s-1] Black Sea Meridional Overturning Circulation strength ocean_meridional_overturning_streamfunction

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: "How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"

5) REFERENCES

Quality Information Document

https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BLK-QUID-OMI_CIRCULATION_MOC_BLKSEA.pdf
https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BLK-PUM-007-004.pdf;
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